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ACADEMIC REFLECTIVE JOURNAL

L03 MACHINE LEARNING WORKFLOW AND LEARNING TYPES

1. INTRODUCTION-(JENNIFER)

In this module 3 lab, L03 ML (Machine Learning) Workflow and Learning Types, it taught me how a workflow can be used to grow and develop the models while also learning to evaluate the data in the model. As a first learning starter from the last lab, introductory of machine learning using algorithms and tools to add to create models with the data given. With this lab of wine sets however showed how to completely explore the data given, to understand the problems given at hand, while picking and using the features to help evaluate performance and being able to interpret the results accurately rather than in predictive route.

Also with learning in this lab of wine classification project, there were some new challenges and terms/tools I never heard of again, which changed my thoughts of machine learning than the idea I had from the last lab. It showed there was more in machine learning by learning and recognizing the differences in reinforcement learning, supervised learning, and unsupervised learning while learning to implement in real world situations or problems in different fields.

2. DESCRIPTION OF EXPERIENCE/TOPIC -(ZAYAAN)

This assignment introduced and helped me understand the three main types of machine learning: supervised, unsupervised, and reinforcement learning. It helped give a more cleared view that each type is used for different kinds of problems and data. Wine Classification project was a great hands-on experience with supervised learning, where a model we train, uses labeled data to learn how to classify different types of wine. I also learned that different machine learning models can produce different results, so comparing their performance is important when deciding which model works best.

3. PERSONAL REFLECTION – (MARTIN)

After I reviewed Module 3, I learned more about machine learning and how it works. I learned about supervised learning, unsupervised learning, and reinforcement learning. I understand that each one can be used for different problems.

I also learned about the Wine Classification project. I reviewed how the data was used to train and test the models. I learned that it is important to check the data and compare different models to see which one works better.

This module helped me understand that machine learning is not only about making a model. We also need to test the model and check the results. I think this will help me understand more about machine learning in the future.

4. DISCUSSION OF IMPROVEMENT AND LEARNING-(ZAYAAN)

I got to practice working on a dataset, where I used machine learning algorithms for classification, and compared their results. One challenge I faced was trying to understand why some models worked better than others, and what could be done to make improvements, which made me realize & understand that machine learning is all about experimenting with various algorithms. At first, understanding the differences between supervised, unsupervised, and reinforcement learning was a little confusing at first but going more through the lab and working through the Wine Classification project made the differences easier to understand. I also noticed that choosing a model is not just about getting an answer but looking at how well the model performs.

5. CONCLUSION – (JENNIFER)

This lab gave a standard and insight on how to improve in the machine learning exercises in future labs and in our daily world events in our lives. This also helps let us know how to have the ability to use multiple tools and know the correct features to use for specific problems that we run into. Being able to also learn concepts with the wine classification lab we learn the new terms that are involved in machine learning which are reinforcement, supervised, and unsupervised into the models created by data and being able to evaluate the result.

6. REFERENCES – (JENNIFER)

For references, I mainly refer to the reference guide to google collab again in case if I get lost or need to do something different. I also use google in case I don't understand any new terms I did not understand or certain tools being used along the way. Lastly, I refer to module 3 powerpoint, to understand what I will run into in this modules lab.